



The auto-search feature has been disabled based on user feedback. Enter a search term/phrase and click "Search" to begin.

[Back to Results](#)

The effect of multiple fixed slots and a trailing-edge flap on the lift and drag of a Clark Y airfoil

Lift and drag tests were made on a Clark Y wing equipped with four fixed slots and a trailing-edge flap in the 5-foot vertical wind tunnel of the National Advisory Committee for Aeronautics. All possible combinations of the four slots were tested with the flap neutral and the most promising combinations were tested with the flap down 45 degrees. Considering both the maximum lift coefficient and the speed-range ratio with the flap neutral no appreciable improvement was found with the use of more than the single leading-edge slot. With the flap down 45 degrees a maximum lift coefficient of 2.60 was obtained but the particular slot combination used had a rather large minimum drag coefficient with the flap neutral. With the flap down 45 degrees the optimum combination, considering both the maximum lift coefficient and the speed-range ratio, was obtained with only the two rearmost slots in use. For this arrangement the maximum lift coefficient was 2.44.

Document ID 19930091501

Acquisition Source Legacy CDMS

Document Type Other

Authors [Weick, Fred E](#)
[Shortal, Joseph A](#)

Date Acquired September 6, 2013

Publication Date January 1, 1933

Report/Patent Number [NACA-TR-427](#) [Report Number: NACA-TR-427](#)

Accession Number 93R20791

Distribution Limits Public

Copyright Work of the US Gov. Public Use Permitted.

Available Downloads

Name	Type	
19930091501.pdf	STI	Download Copy View

Related Records

There are no records associated with this record.



No Preview Available



[No Fear Act](#) | [Freedom of Information Act](#) | [Office of the Inspector General](#) | [Agency Financial Reports](#)

[NASA Web Privacy Policy and Important Notices](#) | [NASA Guidelines for Quality of Information](#)

[Disclaimers, Copyright, Terms of Use](#) | [Accessibility](#) | [NASA OCIO](#) | [USA.gov](#) | [NASA.gov](#)

[Contact Us](#) | [Site Map](#)



NASA Official: JoAnne R. Calhoun

Site Curator: STI Compliance and Distribution Services

Last Modified: June 9, 2026